

Evaluating XAI Metrics in Medical AI: A Cross-Study Co-occurrence Analysis and Evaluation Taxonomy

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Abstract

Explainable Artificial Intelligence (XAI) is increasingly required in clinical settings, yet the metrics used to evaluate explanation quality remain fragmented and rarely compared across medical sub-domains. This paper presents, to our knowledge, the first systematic cross-domain co-occurrence analysis of XAI evaluation metrics in medical AI, examining 52 peer-reviewed studies spanning oncology, cardiology, neuroimaging, radiology, and clinical decision support. We employ a four-phase methodology: systematic corpus collection via PubMed, Scopus, IEEE Xplore, and Google Scholar; criterion extraction and normalisation; cross-domain pairwise correlation analysis; and taxonomy construction. Six quantitative metrics are identified. Faithfulness appears in 47/52 studies (90.4%), with its absence concentrated in human-centred evaluation designs. Sensitivity co-occurs with Faithfulness in 33 studies (63.5%). Among perturbation-based metrics, AOPC/MoRF and Pixel Flipping exhibit the strongest correlation ($\rho = 0.51$, $p < 0.001$), indicating partial methodological redundancy. Sufficiency remains isolated (4/52; 7.7%) despite direct relevance to clinical decision adequacy. Mapping the corpus onto a three-layer taxonomy—mathematical faithfulness, system robustness, and human-centred trust—reveals evaluation effort concentrated in Layer 1 (90.4%), with robustness (34.6%) and human-centred criteria (13.5%) substantially underrepresented. We propose an empirically grounded minimal evaluation framework spanning all three layers and contribute a structured taxonomy validated against cross-domain usage patterns.

1 Introduction

Explainable Artificial Intelligence (XAI) encompasses a set of methods designed to render machine learning models interpretable by revealing internal mechanisms and the reasoning processes underlying their decisions [1]. In clinical medicine, interpretability is not merely desirable but increasingly mandated: clinical decision-making carries direct consequences for patient safety, and regulatory

frameworks across multiple jurisdictions now require that AI-assisted diagnoses be accompanied by interpretable justifications [2]. Despite this regulatory momentum, the metrics used to evaluate explanation quality in medical AI remain poorly standardised. Current evaluation methodologies rely heavily on subjective user studies which, while valuable for assessing clinician perception, are susceptible to confirmation bias and frequently fail to assess the technical fidelity of the generated explanations [3].

This standardisation deficit is compounded by the prevalence of post-hoc explanation methods in clinical AI. Because such explanations are generated after model training, they may not faithfully reflect the model’s actual decision-making logic. In medical contexts, this gap is particularly consequential: a persuasive but unfaithful explanation could reinforce a clinician’s incorrect judgement rather than correct it, potentially compromising patient outcomes.

The present study focuses on the medical domain and addresses three research questions: (RQ1) Which quantitative evaluation metrics are currently employed to assess XAI explanation quality in medical AI? (RQ2) How do these metrics co-occur across studies, and which pairs exhibit potential redundancy? (RQ3) Where do systematic coverage gaps persist across medical sub-domains? By analysing a focused corpus of 52 peer-reviewed medical XAI studies spanning five sub-domains—oncology, cardiology, neuroimaging, radiology, and clinical decision support—we quantify metric co-occurrence patterns, identify redundancies in current practice, and propose a minimal but comprehensive evaluation framework grounded in a three-layer taxonomy of mathematical faithfulness, system robustness, and human-centred trust.

The principal contributions of this paper are threefold:

1. A systematic cross-study co-occurrence analysis quantifying the distribution, pairing patterns, and correlations of XAI evaluation metrics across five medical sub-domains.
2. A three-layer evaluation taxonomy—mathematical faithfulness, system robustness, and human-centred

trust—grounded in empirical usage patterns and validated against the 52-study corpus.

3. A minimal evaluation framework specifying one metric per evaluative dimension, designed to ensure comprehensive coverage without imposing excessive methodological burden.

2 Background

Artificial intelligence is being rapidly integrated into clinical workflows across diagnostic imaging, prognostic modelling, and therapeutic decision support. However, because complex AI models frequently operate as opaque systems, clinicians cannot readily ascertain how diagnostic or prognostic outputs are generated. This opacity creates a pressing need for XAI methods that render medical AI systems transparent and interpretable [4]. XAI has been applied to domains including clinical decision support [5], breast cancer diagnosis [6], cardiovascular risk prediction [7], and responsible cardiovascular disease screening [8].

A central distinction in XAI is between models that are interpretable by design (intrinsically interpretable models) and methods that generate explanations after training (post-hoc methods). Post-hoc approaches are appealing because they can be applied to virtually any deployed clinical system without requiring architectural modification. However, this flexibility entails a fundamental limitation: there is no guarantee that the explanation accurately represents the model’s internal computations, and this fidelity gap remains one of the core unresolved challenges in medical XAI [9, 10].

Two broad paradigms have emerged for evaluating explanation quality in medicine. *Technical evaluation* employs quantitative metrics—such as faithfulness, sensitivity, robustness, and consistency—to assess how accurately an explanation approximates actual model behaviour. *Human-centred evaluation*, by contrast, focuses on the end user, measuring constructs such as clinician trust, cognitive load, and practical usefulness through observational studies and behavioural experiments [11].

While both paradigms have matured considerably, they remain largely disconnected in medical practice. A method that achieves high scores on technical metrics may produce explanations that are opaque or misleading to clinicians, while an explanation that non-specialists find intuitive may lack mathematical fidelity to the underlying model. This disconnect impedes the establishment of shared evaluation standards and limits the comparability of results across medical sub-domains—a problem this paper directly addresses through cross-domain co-occurrence analysis and structured taxonomy construction.

3 Related Work

Efforts to taxonomise XAI methods have produced a substantial but fragmented body of work. Arrieta et al. [1] established the foundational distinction between interpretable-by-design models and post-hoc explanations. Schwalbe and Finzel [12] synthesised over fifty surveys into a framework showing that properties such as faithfulness and stability are widely referenced but inconsistently operationalised. Speith [13] identified a structural source of this inconsistency: taxonomies grounded in mechanisms versus outputs classify the same methods differently, limiting cross-study comparability.

Medical XAI taxonomies address this gap but remain incomplete. Jin et al. [2] distinguish technical faithfulness from clinical usefulness; Holzinger et al. [9] emphasise robustness under distributional shift as unresolved for clinical deployment; Tjoa and Guan [4] survey XAI methods broadly without formalising evaluation criteria. Varghese et al. [14] rank XAI metrics using Borda count and multiple correlation techniques, identifying faithfulness and sensitivity as primary evaluative axes without cross-domain validation. Reviews in radiology and nuclear medicine [15–17] further demonstrate that evaluation metrics are inconsistently reported and rarely subjected to comparative analysis. Reyes et al. [18] argue that explanations must align with clinical reasoning—a requirement frequently absent from benchmark-driven evaluations.

The metrics literature exposes practical consequences of this fragmentation. Rosenfeld [19] demonstrates that most evaluations prioritise algorithmic fidelity while neglecting human-centred dimensions. Kadir et al. [20] confirm that sensitivity, faithfulness, and robustness dominate, with limited attention to human factors. Phuong et al. [21] and Klein et al. [22] provide evidence that without structured taxonomies, metric selection remains ad hoc. Proposed frameworks such as Mirzaei et al. [3] offer selection criteria but lack domain-specific grounding. Complementary analyses [23, 24] reinforce that the absence of shared evaluation vocabulary impedes meaningful synthesis.

Human-centred evaluation has received increasing attention. Chen et al. [25] identify the systematic absence of human-centred components in medical XAI pipelines. Bauer and Michalowski [26] highlight cognitive load, trust calibration, and decision accuracy as key constructs requiring operationalisation. Dey et al. [27] document continued methodological fragmentation, and Müller et al. [28] emphasise that regulatory demands for explainability and causability extend beyond standard faithfulness metrics. Recent surveys [29–31] confirm that standardised evaluation frameworks remain underdeveloped, with Ahmed et al. [31] identifying metric standardisation as a major open challenge.

What remains absent is a unified taxonomy grounded in empirical usage patterns rather than purely conceptual design. No existing framework jointly spans oncology, cardiology, neuroimaging, radiology, and clinical decision support while integrating mathematical faithfulness, system robustness, and human-centred trust. The present study addresses this gap through systematic co-occurrence analysis and empirically validated taxonomy construction.

4 Methodology

To systematically characterise how XAI evaluation metrics are employed within medical AI, we adopt a four-phase methodology. Rather than treating each sub-domain as isolated, we extract, normalise, and align evaluation criteria across oncology, cardiology, neuroimaging, radiology, and clinical decision support into a single coherent corpus, then analyse metric co-occurrence and correlation patterns.

4.1 Phase 1: Systematic Corpus Collection

Databases and Search Strategy. We conducted a systematic literature search across four databases: PubMed, Scopus, IEEE Xplore, and Google Scholar. Representative query strings included:

- ("explainable AI" OR "XAI") AND ("evaluation" OR "metrics") AND ("medical imaging" OR "clinical")
- ("saliency" OR "attribution") AND ("evaluation" OR "faithfulness") AND ("healthcare" OR "radiology")
- ("explainability" OR "interpretability") AND ("clinical decision support") AND ("quantitative evaluation")

Results were restricted primarily to 2020–2025, with seminal earlier works retained where they provided foundational evaluation methodologies. Searches were last updated in January 2026.

Inclusion and Exclusion Criteria. Studies were included if they: (I1) were peer-reviewed in a Q2+ venue (Scopus/Scimago) or recognised conference (IEEE, ACM, IJCAI, MICCAI), with limited Q3 exceptions; (I2) addressed at least one of the five medical sub-domains; and (I3) contained explicit evaluation of XAI quality using at least one quantitative metric or structured human-centred assessment. Studies were excluded if they: (E1) addressed non-medical AI exclusively; (E2) proposed XAI methods without quantitative evaluation; (E3) were duplicates; or (E4) were published below Q3 without novel domain-specific methodologies. A

limited number of Q3 articles were included where they provided domain-specific evaluation methodologies not available in higher-ranked venues.

Screening and Selection Process. Two independent reviewers conducted screening. An initial search yielded 327 candidate records. After removing 89 duplicates, 238 unique records were screened by title and abstract; 143 were excluded for lacking relevance. Disagreements were resolved through discussion and, where necessary, full-text consultation. The remaining 95 full-text articles were assessed against inclusion/exclusion criteria; 43 were excluded for not reporting quantitative XAI evaluation metrics or falling outside defined sub-domains. Inter-rater agreement was substantial (Cohen’s $\kappa = 0.82$). The final corpus comprises 52 peer-reviewed evaluation studies.

Study Selection Flow. Following PRISMA reporting conventions [32]:

1. **Identification:** 327 records (PubMed: 78; Scopus: 112; IEEE Xplore: 84; Google Scholar: 53).
2. **Screening:** 89 duplicates removed $\rightarrow$ 238 unique records. Title/abstract screening excluded 143.
3. **Eligibility:** 95 full-text articles assessed. 43 excluded (no quantitative XAI metrics: $n = 28$; outside sub-domains: $n = 11$; below venue threshold: $n = 4$).
4. **Inclusion:** 52 studies in the final corpus.

Corpus Justification. The corpus of $n = 52$ studies, while modest in absolute size, provides sufficient coverage for the analytical objectives of this study. The corpus spans five distinct medical sub-domains (oncology, cardiology, neuroimaging, radiology, and clinical decision support) with representation from 28 unique venues ranging from top-tier journals (Nature Machine Intelligence, Medical Image Analysis, IEEE TMI) to specialised clinical outlets. This cross-domain and cross-venue breadth ensures that observed metric usage patterns are not artefacts of a single community’s conventions. Moreover, the corpus captures both technically oriented and human-centred evaluation designs, enabling the taxonomic analysis reported below. For binary co-occurrence analysis involving six metrics, a sample of 52 studies yields 15 pairwise comparisons with adequate statistical power to detect moderate-to-strong associations ($\rho \geq 0.35$) at $\alpha = 0.05$.

The corpus incorporates domain-specific evaluation works including clinician-grounded studies in cardiology and clinical decision support [33–37], imaging-focused analyses [10, 38–41], neuroimaging studies [42–44], and human-centred evaluation studies [45, 46].

4.2 Phase 2: Criterion Extraction and Normalisation

From each study, we recorded every evaluation criterion and its operationalisation. The same concept frequently appears under different names—“faithfulness,” “fidelity,” and “completeness” are often used interchangeably, a terminological inconsistency that Speith [13] identifies as a main obstacle to shared evaluation standards.

We address this in two steps. First, each criterion is mapped to a canonical label grounded in established XAI evaluation literature [19, 20]. Second, each measurement approach is classified along two dimensions: (a) quantitative versus user-study-based, and (b) overall model behaviour versus individual predictions. This mirrors the functionally-grounded and human-grounded distinction from Kadir et al. [20]. Varghese et al. [14] further informed our normalisation by applying Borda count and correlation techniques to rank competing metrics, providing empirical justification for treating faithfulness and sensitivity as primary axes. The resulting normalised list comprises six metrics—Faithfulness, Sensitivity, AOPC/MoRF, Pixel Flipping, IROF, and Sufficiency—comparable across sub-domains regardless of original terminology.

4.3 Formal Definitions of Evaluation Metrics

To ensure terminological precision and reproducibility, we formally define each metric. All definitions follow established conventions.

Faithfulness. Faithfulness quantifies how accurately an explanation reflects the model’s decision-making process [47]. For model f , input x , and explanation $\phi(x)$:

$$\text{Faith}(f, x, \phi) = \text{corr}(\phi(x)_i, f(x) - f(x_{\setminus i}))_{i=1}^d \quad (1)$$

where $\phi(x)_i$ is the importance score for feature i , $x_{\setminus i}$ is the input with feature i removed, and d is the feature count.

Sensitivity. Sensitivity quantifies explanation responsiveness to meaningful input changes [47]:

$$\text{Sens}_{\max}(f, x, \phi) = \max_{\|x' - x\| \leq \epsilon} \|\phi(x') - \phi(x)\| \quad (2)$$

where ϵ defines the perturbation radius. Well-behaved explanations exhibit high sensitivity to prediction-altering changes and low sensitivity otherwise.

AOPC/MoRF. AOPC measures average prediction change as features are progressively removed in order of

decreasing importance [48]:

$$\text{AOPC} = \frac{1}{K+1} \sum_{k=0}^K \left\langle f(x^{(0)}) - f(x_k^{(\text{MoRF})}) \right\rangle_x \quad (3)$$

where $x_k^{(\text{MoRF})}$ is the input after removing the k most relevant features in Most-Relevant-First order.

Pixel Flipping. Pixel Flipping systematically replaces input elements in saliency-map order [48]:

$$\text{PF}(k) = f(x^{(0)}) - f(\tilde{x}_k) \quad (4)$$

where $\tilde{x}_k$ denotes the input after flipping the top- k pixels. Pixel Flipping is mechanistically related to AOPC but operates at pixel-level granularity.

IROF. IROF extends perturbation-based evaluation by removing semantically meaningful segments [49]:

$$\text{IROF} = \frac{1}{S} \sum_{s=1}^S [f(x) - f(x_{\setminus s})] \quad (5)$$

where $x_{\setminus s}$ denotes the input with segment s removed. IROF is particularly suited to medical imaging where anatomical structures correspond to contiguous pixel regions.

Sufficiency. Sufficiency measures whether identified features alone reproduce the model’s prediction [47]:

$$\text{Suff}(f, x, \phi) = f(x) - f(x_\phi) \quad (6)$$

where x_ϕ retains only the features identified as important. A score near zero indicates that the explanation captures the minimal information necessary for the decision. Sufficiency is conceptually distinct from consistency (similar inputs receiving similar explanations) and from faithfulness (correlation between attributed and actual importance). Sufficiency directly assesses whether highlighted features alone sustain the original prediction—a property of particular clinical relevance, determining whether the explanation identifies a decision-sufficient evidence set.

4.4 Phase 3: Cross-Domain Alignment and Correlation Analysis

The six normalised metrics were compared across the five sub-domains in a comparison matrix (rows: criteria; columns: sub-domains). Criteria appearing in ≥ 3 sub-domains are classified as *cross-domain*; those in 1–2 as *domain-specific*. This classification reflects Schwalbe and Finzel’s [12] observation that properties such as faithfulness recur broadly but with divergent operationalisations.

Statistical Approach. To quantify co-occurrence and potential redundancy, we compute **Spearman’s rank correlation** (ρ) across all 15 pairwise metric combinations using the full 52-study corpus. Each metric is encoded as a binary variable (presence = 1, absence = 0). In this binary encoding, Spearman’s ρ is mathematically equivalent to the phi coefficient (ϕ), a standard measure of association for dichotomous variables. We selected Spearman’s ρ (equivalently ϕ) because: (i) it imposes no distributional assumptions; (ii) it is interpretable as monotonic association for binary indicators; and (iii) it facilitates comparison with prior rank-based correlation studies in XAI evaluation.

Two limitations of binary correlation should be noted. First, binary encoding discards information about the intensity or quality of metric application. Second, high co-occurrence does not establish construct equivalence; it may reflect convention or methodological convenience. These limitations are addressed in Section 6.3 (Threats to Validity).

Where the same criterion was measured differently across sub-domains, we compared operationalisations and recorded which approach is more theoretically grounded. This phase also surfaces genuine tensions—cases where an explanation achieves high technical scores but is judged unclear by clinicians—taken as evidence that explanation quality is multidimensional.

4.5 Phase 4: Taxonomy Construction

The final phase organises aligned criteria into a three-layer structure, building on Kadir et al. [20] while adding an explicit robustness dimension that prior medical XAI taxonomies have addressed inconsistently [2, 13].

The choice of three layers reflects both theoretical grounding and empirical observation. The distinction between technical fidelity (Layer 1) and user-facing evaluation (Layer 3) is well established [2, 20]. The addition of a robustness layer (Layer 2) is motivated by: first, Holzinger et al. [9] identifying robustness under distributional shift as unresolved in medical AI; second, our corpus analysis revealing that robustness-oriented evaluation constitutes a coherent cluster that does not reduce to either faithfulness measurement or clinician-facing assessment. This structure is consistent with Speith [13] and the faithfulness–usefulness separation of Jin et al. [2].

During construction, we applied a primarily deductive approach, assigning metrics to layers based on formal definitions and alignment with prior frameworks. Two inductive adjustments were made: (i) Sufficiency was assigned to Layer 1 on the grounds that its operationalisation (Equation 6) is purely computational, despite its clinical relevance; (ii) contrastive explanations were flagged as cross-layer.

Layer 1 (Mathematical Faithfulness) covers

whether the explanation accurately reflects model computation, including faithfulness correlation, sufficiency, sensitivity, and fidelity under perturbation [2, 20].

Layer 2 (System Robustness) addresses whether explanations remain stable when patient inputs shift, models are updated, or adversarial conditions arise. Repetto et al. [39] provide direct evidence that XAI explanations degrade under both natural and adversarial data corruption.

Layer 3 (Human-Centred Trust) covers whether explanations help clinicians understand, decide, and calibrate confidence. Speith [13] argues that most taxonomies neglect this dimension; in clinical contexts, a technically faithful but opaque explanation provides limited practical value.

Cross-layer criteria (e.g., contrastive explanations) are flagged to maintain transparency about inter-layer connections rather than enforcing artificial boundaries. The resulting framework functions as both a descriptive map of current evaluation practice and a diagnostic tool for identifying coverage gaps.

5 Results

The following sections report findings from the four-phase methodology applied to the 52-study corpus, organised by corpus composition, metric co-occurrence, cross-domain alignment, and taxonomy validation. Interpretive analysis is deferred to the Discussion (Section 6).

5.1 Corpus Composition and Metric Distribution

The corpus spans 52 evaluation studies across five medical sub-domains: oncology ($n = 9$), radiology ($n = 12$), cardiology ($n = 9$), neuroimaging ($n = 7$), and clinical decision support ($n = 15$).¹ Table 1 provides a structured summary.

Six quantitative evaluation metrics were identified: Faithfulness, Sensitivity, AOPC/MoRF, Pixel Flipping, IROF, and Sufficiency. Figure 1 presents their distribution. Faithfulness was the most frequently reported metric (47/52; 90.4%), followed by Sensitivity (34/52; 65.4%), Pixel Flipping (14/52; 26.9%), AOPC/MoRF (13/52; 25.0%), and IROF and Sufficiency (each 4/52; 7.7%).

¹Studies categorised under Clinical AI, Clinical ML, Healthcare ML, and Medical Imaging are grouped under “Clinical” for cross-domain comparison, consolidating studies that address general clinical evaluation methodology or span multiple imaging modalities.

Table 1: Evaluation studies included in the corpus ($N = 52$), Part 1: Cardiology, Oncology, and Radiology. Domains: Card. = Cardiology; Onc. = Oncology; Rad. = Radiology; Neuro. = Neuroimaging; Clin. = Clinical.

Study	Domain	Venue	Notes
[50]	Card.	iScience	Perturbation-based saliency for 12-lead ECG
[35]	Card.	J. Clin. Med.	Grad-CAM sensitivity in prenatal echocardiography
[33]	Card.	Comput. Biol. Med.	SHAP inspection for heart-failure mortality
[34]	Card.	JMIR Med. Inform.	Interpretable HCM progression modelling
[36]	Card.	Front. Cardiovasc. Med.	Attribution stability for HF survival
[51]	Card.	FLAIRS	Conceptual explanation for ECG classification
[52]	Card.	Symmetry	Feature-level CHF explanation
[7]	Card.	Sci. Rep.	SHAP-based CVD risk prediction
[8]	Card.	Sci. Rep.	XAI chatbot for CVD screening
[40]	Onc.	Sci. Rep.	Saliency vs. lesion ground truth in breast cancer
[6]	Onc.	Sci. Rep.	Lesion-focused DL for breast cancer
[53]	Onc.	IEEE Access	SHAP/Grad-CAM bias analysis in breast cancer
[5]	Onc.	Healthcare	XAI meta-analysis for clinical decision support
[41]	Onc.	Comput. Meth. Prog. Biomed.	AOPC + Max-Sensitivity for attribution maps
[54]	Onc.	Pattern Recognit.	Multi-task attention for COVID-19 CXR
[55]	Onc.	Eur. J. Radiol.	Quantitative saliency evaluation in mammography
[56]	Onc.	Int. J. Intell. Syst.	Fuzzy ensemble for cervical cancer
[38]	Rad.	Sci. Rep.	Heatmap localisation for cardiomegaly CXR
[57]	Rad.	J. Pers. Med.	Scalability evaluation for COVID-19 CXR
[58]	Rad.	IEEE TMI	Domain-aware interpretation; misleading attributions
[59]	Rad.	IEEE Access	Example-based XAI review in medical imaging
[10]	Rad.	Sensors	Bias-sensitive saliency for COVID CT
[39]	Rad.	Mach. Learn.	Robustness under corruption in medical images
[45]	Rad.	PLoS ONE	Human-centred Grad-CAM vs LIME study
[60]	Rad.	Nat. Mach. Intell.	Saliency benchmarking for CXR
[61]	Rad.	Radiology: AI	Saliency trustworthiness for abnormality detection
[62]	Rad.	Radiology: AI	Attention saliency for pneumothorax
[63]	Rad.	Sci. Rep.	Patch perturbation for COVID-19 CXR
[64]	Rad.	BioData Mining	Visual explainability with statistical analysis

Table 1 (continued): Neuroimaging and Clinical sub-domains.

Study	Domain	Venue	Notes
[43]	Neuro.	Sensors	Saliency sensitivity for brain tumour
[42]	Neuro.	Heliyon	XAI-augmented CNN for brain tumour
[65]	Neuro.	Med. Image Anal.	Counterfactual explanation evaluation
[44]	Neuro.	WIREs DMKD	XAI review in intracranial haemorrhage
[11]	Neuro.	MICCAI	Counterfactual impact analysis
[66]	Neuro.	SPIE ICBBE	Consistency benchmarking for brain MRI
[67]	Neuro.	Med. Eng. Phys.	LIME/Grad-CAM/SHAP for brain tumour
[68]	Clin.	IEEE J-BHI	Trustworthiness metrics for IoMT imaging
[46]	Clin.	IEEE Access	Co-design with radiologists for human-centred trust
[69]	Clin.	Artif. Intell. Med.	Explanation validation in CAD diagnosis
[37]	Clin.	IEEE PerCom Wksp.	qxAI faithfulness + sensitivity metrics
[70]	Clin.	Sci. Rep.	Domain-aware criteria beyond predictive metrics
[71]	Clin.	Int. J. HCI	Explainability effects on clinician trust
[72]	Clin.	KI-Künstl. Intell.	Human-centred lessons for medical decisions
[26]	Clin.	JAMIA	Human-centred explainability review
[73]	Clin.	Appl. Sci.	Grad-CAM usefulness across modalities
[2]	Clin.	Med. Image Anal.	Clinical evaluation guidelines
[74]	Clin.	WIREs Mech. Dis.	Attribution XAI survey in healthcare
[75]	Clin.	ACM Comput. Surv.	XAI survey with metrics benchmarking
[76]	Clin.	AAAI	Multi-modal XAI against clinical requirements
[77]	Clin.	Comput. Meth. Prog. Biomed.	Systematic faithfulness evaluation of DL methods
[78]	Clin.	Comput. Intell.	XAI experiments on infrared breast images
[79]	Clin.	Comput. Meth. Prog. Biomed.	Perturbation-based XAI for wound healing

The five studies lacking Faithfulness comprise three human-centred evaluation designs [26, 71, 72] and two consistency-focused benchmarks [56, 66]. This concentration pattern is addressed in the Discussion.

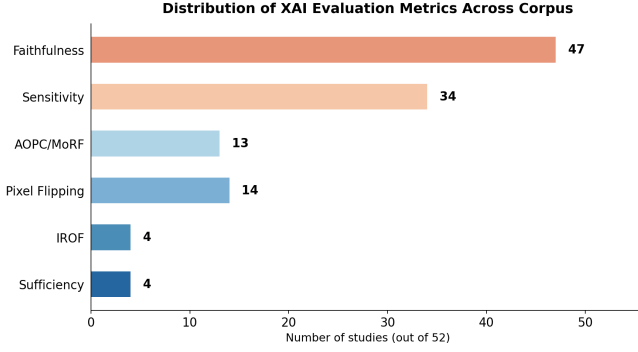


Figure 1: Distribution of XAI evaluation metrics across the 52-study corpus. Faithfulness is the most prevalent (47/52; 90.4%), while IROF and Sufficiency are each reported in only 4 studies (7.7%).

5.2 Metric Co-occurrence and Correlation

Pairwise co-occurrence analysis revealed three distinct patterns, illustrated in Figures 3 and 2.

First, Faithfulness and Sensitivity co-occurred in 33 of the 47 Faithfulness-reporting studies (70.2%; 63.5% of the full corpus). In the five studies lacking Faithfulness, Sensitivity appeared in one case [66]. Table 2 lists studies in which sensitivity analysis formed a primary evaluation method.

Second, AOPC/MoRF and Pixel Flipping co-occurred in 7 studies and produced the strongest Spearman correlation ($\rho = 0.51$, $p < 0.001$). AOPC/MoRF and IROF also exhibited significant correlation ($\rho = 0.37$, $p = 0.007$). The Faithfulness–Sensitivity correlation was modest but significant ($\rho = 0.33$, $p = 0.016$).

Third, Sufficiency exhibited negligible co-occurrence with all other metrics, producing negative correlations ranging from $\rho = -0.09$ to $\rho = -0.15$ (all non-significant at $\alpha = 0.05$).

5.3 Cross-Domain Alignment Within Medical Sub-domains

Comparison of the normalised criterion list across sub-domains revealed both cross-domain and domain-specific patterns. Faithfulness and Sensitivity qualified as near-cross-domain criteria, appearing in ≥ 4 sub-domains with broadly comparable operationalisations. Table 3 summarises studies explicitly reporting Faithfulness Correlation or Faithfulness Estimate as their primary evaluation metric.

Table 2: Representative studies using sensitivity or perturbation-based robustness checks.

Reference	Method	Application
[50]	Guided saliency + perturbation	Arrhythmia detection
[35]	Grad-CAM sensitivity	Prenatal echo.
[38]	Heatmap localisation checks	Cardiomegaly
[36]	Attribution stability	HF survival
[43]	Saliency sensitivity	Brain tumour
[40]	Saliency vs. lesion masks	Breast cancer
[39]	Corruption robustness	Medical imaging
[41]	AOPC + Max-Sensitivity	Dermatol./histopath.
[37]	qxAI sensitivity metrics	Cardiac disease
[61]	Localisation evaluation	Radiology
[60]	Saliency benchmarking	CXR interpretation
[62]	Attention vs. gradient	Pneumothorax
[63]	Patch perturbation	COVID-19 CXR
[55]	Quantitative saliency	Mammography
[66]	Consistency benchmarking	Brain MRI
[64]	Statistical saliency analysis	Medical imaging
[79]	Cross-corpus perturbation	Wound healing

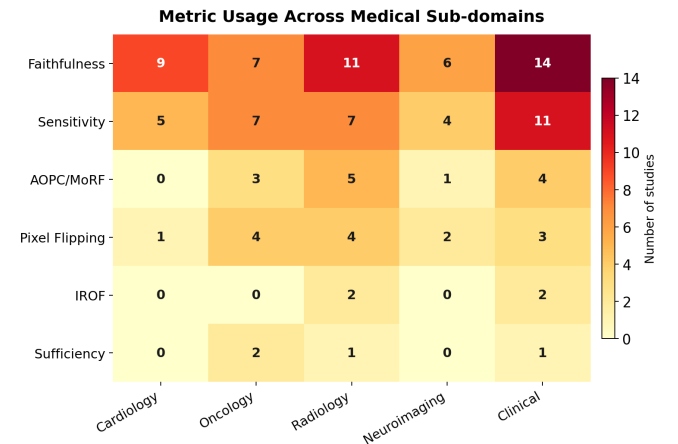


Figure 2: Metric usage across medical sub-domains. Each cell reports study count employing a given metric within a sub-domain. Faithfulness and Sensitivity appear across all five sub-domains; IROF and Sufficiency are concentrated in radiology and clinical settings.

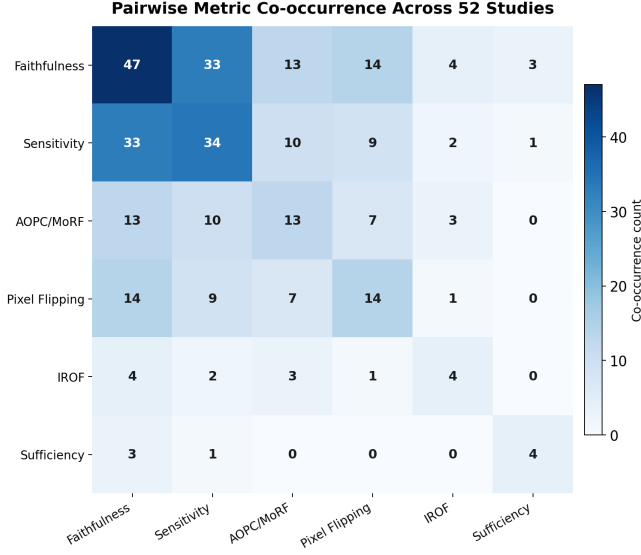


Figure 3: Pairwise metric co-occurrence across 52 studies. Diagonal entries indicate total frequency; off-diagonal entries indicate joint reporting. The Faithfulness–Sensitivity pair shows the highest absolute co-occurrence ($n = 33$).

IROF and Pixel Flipping were predominantly concentrated in radiology and oncology, where saliency-map explanations are standard. The radiology sub-corpus—including benchmarking studies such as Saporta et al. [60] and Wollek et al. [62]—exhibited the highest metric diversity. Neuroimaging studies relied predominantly on Sensitivity, with notably fewer metrics per study; Qin [66] represents an emerging exception. Oncology studies were the most metrically diverse, combining technical and user-study-based criteria, with Cerekci et al. [55] providing dedicated quantitative evaluation in mammography. The clinical sub-domain expanded most substantially, incorporating human-centred studies [26, 71, 72] that operationalise Layer 3 trust criteria.

5.4 Taxonomy Validation Against the Corpus

Mapping the 52 studies onto the three-layer taxonomy yielded the following distribution: 47/52 studies (90.4%) reported at least one mathematical faithfulness criterion (Layer 1); 18 (34.6%) addressed system robustness through multi-metric perturbation analyses (Layer 2); and 7 (13.5%) included any human-centred trust measure or sufficiency check (Layer 3). Figure 4 illustrates the taxonomy structure and empirical distribution.

Ihongbe and Fouad [45], Fouad et al. [46], Rezaeian et al. [71], Bauer and Michalowski [26], and Finzel [72] represent the most explicit operationalisations of the trust layer, collectively demonstrating that cognitive load, clinician comprehension, and diagnostic confidence are measurable constructs.

Table 4 lists perturbation-based and pixel-flipping

Table 3: Representative studies reporting faithfulness, fidelity, or domain-aware explanation validity.

Reference	Method	Application
[33]	SHAP inspection	HF mortality
[34]	Interpretable progression	HCM
[52]	Feature-level explanation	CHF
[42]	Saliency faithfulness	Brain tumour
[6]	Lesion-focused detection	Breast cancer
[11]	Counterfactual impact	Medical imaging
[58]	Domain-aware validity	Medical imaging
[68]	Trustworthiness criteria	IoMT imaging
[37]	Faithfulness correlation	Cardiac disease
[76]	Multi-modal evaluation	Medical imaging
[77]	Systematic faithfulness	Medical imaging
[41]	AOPC + Max-Sensitivity	Image classification
[73]	Grad-CAM assessment	Multi-modality
[55]	Saliency faithfulness	Mammography
[78]	Faithfulness experiments	IR breast imaging

studies, representing the most common operationalisation of the faithfulness layer in imaging contexts. Cross-layer criteria—most notably contrastive explanations—appeared in several studies and were flagged accordingly.

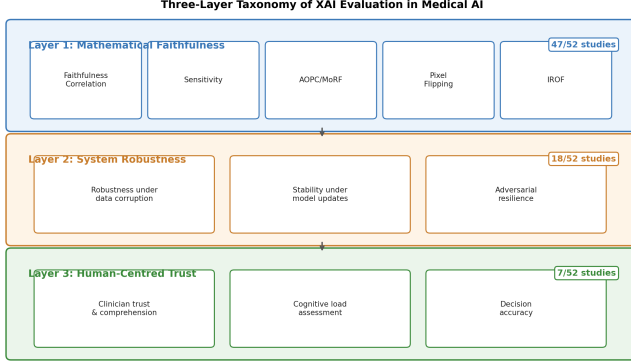


Figure 4: Three-layer taxonomy of XAI evaluation in medical AI. Layer 1 (Mathematical Faithfulness) is addressed by 47/52 studies (90.4%), Layer 2 (System Robustness) by 18/52 (34.6%), and Layer 3 (Human-Centred Trust) by 7/52 (13.5%).

Table 4: Representative perturbation, counterfactual, and corruption-based evaluation studies.

Reference	Method	Application
[50]	Input perturbation + attribution	ECG diagnosis
[36]	Feature ablation	HF survival
[65]	Counterfactual generation	Medical imaging
[11]	Counterfactual impact	Image interpretation
[10]	Bias-sensitive saliency	COVID CT
[40]	Saliency-mask comparison	Breast lesion
[39]	Corruption robustness	Medical imaging
[41]	AOPC/MoRF perturbation	Image classification
[63]	Patch perturbation	COVID-19 CXR
[79]	Cross-corpus perturbation	Wound healing
[67]	Multi-method perturbation	Brain tumour
[56]	Reliability-weighted perturbation	Cervical cancer

6 Discussion

The results reveal both encouraging convergences and persistent gaps in medical XAI evaluation. This section interprets key findings, analyses metric redundancy, proposes a minimal evaluation framework, discusses clinical implications, and identifies threats to validity.

6.1 Interpretation of Key Findings

Expanding the corpus to 52 studies introduces an important revision to the Faithfulness finding. Rather than appearing universally, Faithfulness was present in 47/52 studies (90.4%), with the five exceptions concentrated in human-centred and consistency-focused designs [26,56,66,71,72]. This pattern is analytically significant: Faithfulness absence is not randomly distributed but co-occurs with a shift in evaluation layer, indicating a genuine conceptual divide between technical-fidelity and clinician-trust research paradigms. Studies prioritising clinician-facing evaluation operate under a distinct evaluative logic in which mathematical fidelity is not the primary concern—consistent with Rosenfeld’s [19] observation that technical and human-centred evaluation priorities tend to diverge.

Because Faithfulness exhibits non-trivial variance, Spearman correlations involving it are computable. The Faithfulness–Sensitivity correlation ($\rho = 0.33$, $p = 0.016$) confirms co-occurrence without implying full redundancy: both metrics target Layer 1 but assess complementary properties—Faithfulness measures fidelity of attributed importance scores (Equation 1), while Sensitivity measures explanation stability under input perturbation (Equation 2).

6.2 Metric Redundancy and Complementarity

The strongest pairwise correlation is between AOPC/MoRF and Pixel Flipping ($\rho = 0.51$, $p < 0.001$), a result that is mechanistically expected: both are perturbation-based methods that degrade saliency-ranked features and measure the resulting prediction drop. Critically, AOPC operates at the feature level (Equation 3), while Pixel Flipping operates at pixel-level granularity (Equation 4). Their high co-occurrence correlation reflects *methodological overlap*—both instantiate the same perturbation-and-measure paradigm—rather than strict *conceptual redundancy*. In practice, this means that reporting both metrics within a single study adds limited incremental information: selecting one perturbation-based metric (AOPC for feature-level; Pixel Flipping for pixel-level) is typically sufficient.

AOPC/MoRF and IROF also correlate significantly ($\rho = 0.37$, $p = 0.007$). This pair exhibits a different relationship: AOPC removes individual features, whereas IROF removes semantically coherent segments (Equation 5). Their correlation thus reflects a shared perturbation paradigm, but IROF captures contributions of anatomically meaningful regions—making it complementary rather than redundant in medical imaging applications where anatomical structures are clinically relevant.

By contrast, Sufficiency exhibited negative correla-

tions with all other metrics (ρ from -0.09 to -0.15 , all non-significant). This isolation reflects a conceptual gap: Sufficiency asks whether the identified features *alone* suffice to reproduce the prediction (Equation 6), a question that the remaining metrics do not address. The negative correlations suggest that studies employing Sufficiency tend to adopt a fundamentally different evaluation philosophy, further supporting the position that Sufficiency captures a non-redundant evaluative dimension.

Figure 5 presents the full correlation matrix. Figure 6 visualises these relationships as a metric correlation network, highlighting the perturbation-based redundancy cluster and the isolation of Sufficiency.

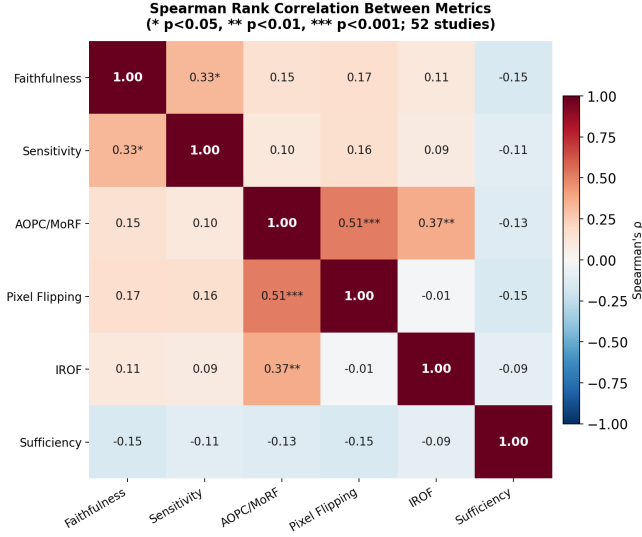


Figure 5: Spearman rank correlation (ρ) between XAI evaluation metrics across the 52-study corpus (equivalent to the phi coefficient for binary data). Statistical significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The AOPC/MoRF–Pixel Flipping pair exhibits the strongest correlation ($\rho = 0.51$), while Sufficiency shows negative correlations with all other metrics.

The sub-domain disparity between oncology and neuroimaging further underscores that metric richness is unevenly distributed. Oncology benefits from well-established imaging benchmarks supporting diverse evaluation protocols, whereas neuroimaging lacks analogous shared resources, constraining both metric diversity and cross-study comparability. Targeted benchmark development for underserved sub-domains could have a disproportionately positive effect on standardisation.

6.3 Toward a Minimal Evaluation Framework

Building on the co-occurrence patterns, correlation analysis, and taxonomy mapping, we propose a minimal evaluation framework that spans all three taxonomy layers while imposing a tractable methodological burden. To our knowledge, this is the first such framework empir-

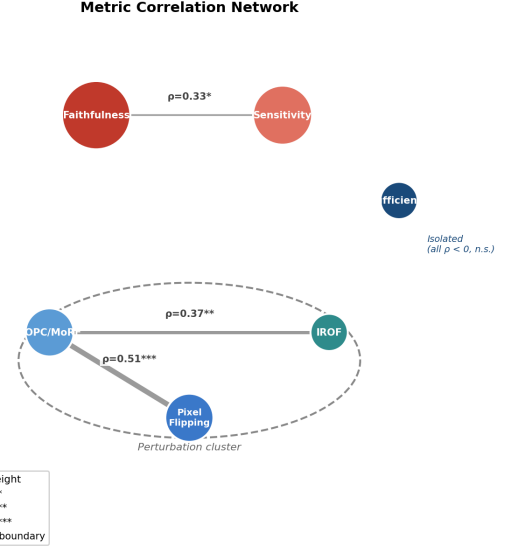


Figure 6: Metric correlation network derived from Spearman’s ρ across the 52-study corpus. Nodes represent evaluation metrics; edge thickness and colour encode correlation strength. The dashed boundary highlights the perturbation-based redundancy cluster (AOPC/MoRF, Pixel Flipping, IROF). Sufficiency is visually isolated, confirming its non-redundant evaluative role. Only statistically significant correlations ($p < 0.05$) are shown as edges.

ically grounded in cross-domain usage patterns rather than purely conceptual design. The framework specifies five evaluation dimensions, each addressed by at least one metric. Table 5 presents the framework in compact form; Figure 7 illustrates the three-layer structure.

This five-component framework ensures coverage of all three taxonomy layers without requiring simultaneous application of all six metrics, several of which exhibit partial redundancy. Researchers operating under resource constraints may prioritise Faithfulness, one perturbation metric, and one human-centred measure as a minimum viable set, provided that the omission of Sensitivity and Sufficiency is explicitly acknowledged and justified.

Table 5: Proposed minimal evaluation framework for medical XAI. Each row specifies one required evaluation dimension, its taxonomy layer, evaluative role, and rationale. † indicates that the metric may be omitted under resource constraints if the omission is explicitly justified.

Metric	Layer	Req.	Rationale
Faithfulness	L1	Req.	Foundation of technical credibility; reported in 90.4% of corpus
Perturbation (AOPC or PF or IROF)	L1	Req.	Validates that removal of attributed features degrades prediction; one metric sufficient given redundancy ($\rho = 0.51$)
Sensitivity	L1/L2	Req.†	Bridges faithfulness and robustness; captures responsiveness to input change
Sufficiency	L1	Req.†	Only metric assessing whether explanation identifies decision-sufficient evidence; currently at 7.7% adoption
Human-centred (trust / cognitive load / decision accuracy)	L3	Req.	Essential for deployment; only 13.5% of corpus addresses this layer

6.4 Clinical Implications

The structural imbalance documented by the taxonomy mapping carries direct clinical consequences. Two omission risks are particularly concerning.

Risk of missing Sufficiency. When Sufficiency is not evaluated, there is no guarantee that the features highlighted by an explanation are, on their own, adequate to sustain the model’s prediction. In clinical practice, this means a clinician may act on an explanation that identifies *correlated* but not *causally sufficient* features—potentially leading to diagnostic decisions based on incomplete evidence. For example, a saliency map highlighting peri-lesion tissue rather than the lesion itself may achieve high Faithfulness scores (the model *does* attend to that region) while failing Sufficiency (the highlighted region alone does not reproduce the prediction). Without Sufficiency evaluation, such failure modes remain undetected.

Risk of missing Robustness. When robustness is not evaluated (absent in 65.4% of corpus studies), explanations may be valid only under the specific data distribution on which they were generated. In multi-site clinical deployment—where imaging equipment, patient demographics, and acquisition protocols vary across hospitals—an explanation that is faithful under training conditions may become unstable or misleading under distributional shift. Repetto et al. [39] provide empirical evidence that XAI explanations degrade meaningfully under both natural and adversarial corruption, confirming that robustness cannot be assumed but must be explicitly tested prior to clinical deployment.

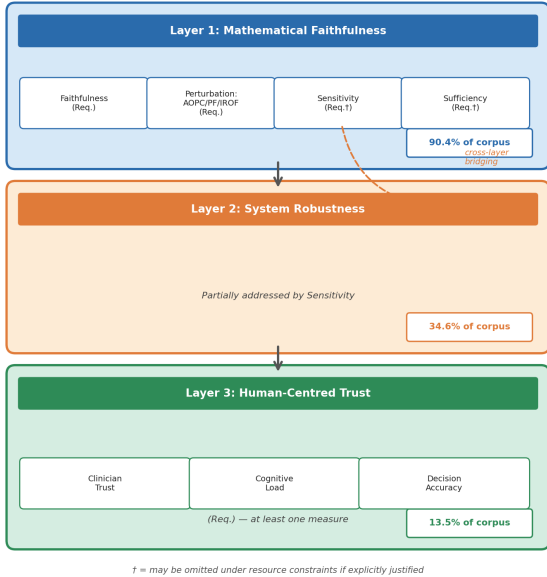


Figure 7: Proposed minimal evaluation framework organised by taxonomy layer. Layer 1 (Mathematical Faithfulness) requires Faithfulness, one perturbation metric, Sensitivity, and Sufficiency. Layer 2 (System Robustness) is partially addressed by Sensitivity. Layer 3 (Human-Centred Trust) requires at least one clinician-facing measure. Arrows indicate cross-layer bridging by Sensitivity.

6.5 Threats to Validity

Internal Validity. Metric classification into canonical categories requires interpretive judgement. Although we followed established normalisation procedures [19,20], different coding decisions could alter metric distribution. Binary encoding discards information about depth or quality of metric application: a study reporting Faithfulness as its central axis is coded identically to one mentioning it peripherally. The screening process, while conducted by two independent reviewers with substantial agreement (Cohen’s $\kappa = 0.82$), is not immune to selection bias.

External Validity. The corpus of 52 studies spans five sub-domains but remains modest in absolute size. Observed patterns should be interpreted as indicative of current trends rather than definitive characterisations, requiring replication on larger samples. The restriction to medical XAI limits generalisability to other high-stakes domains. Evidence from non-medical applications [80,81] suggests similar fragmentation may exist elsewhere.

Construct Validity. Co-occurrence analysis captures which metrics tend to appear together but cannot establish whether co-occurring metrics are genuinely redundant (measuring the same construct) or merely reported together by convention. Spearman correlations measure association in usage patterns, not agreement in evaluation outcomes. Establishing true redundancy would require applying multiple metrics to the same explanation–dataset pairs—a task outside this review’s scope. The taxonomy’s three layers were constructed primarily through deductive reasoning; prospective clinical practitioner studies would be needed to confirm that the trust layer captures what domain experts actually require.

6.6 Limitations

Several additional limitations qualify these findings. First, the corpus size of 52 studies limits statistical power for low-frequency metrics such as IROF and Sufficiency (each $n = 4$); correlations involving these metrics should be interpreted with caution. Second, the study does not include direct clinician validation of the proposed taxonomy or minimal framework; prospective user studies are needed to confirm practical utility. Third, metric redundancy inferred from co-occurrence patterns remains uncertain without head-to-head comparisons on shared datasets. Fourth, the corpus prioritises English-language publications in indexed venues, potentially underrepresenting evaluation practices in non-English research communities.

6.7 Future Work

Three directions follow from these findings. First, a prospective study instrumenting clinicians as they interact with SHAP- and Grad-CAM-generated explanations would ground the trust layer in behavioural evidence. Second, developing a standardised reporting template—specifying the minimal metric set proposed in Section 6.3 alongside optional domain-specific additions—could reduce the documented fragmentation. Third, extending analyses to include user-study metrics such as cognitive load and decision accuracy would enable testing whether high Faithfulness scores translate into genuine clinical utility.

7 Conclusion

This paper investigated the evaluation landscape of XAI within medical AI through a systematically assembled corpus of 52 peer-reviewed studies spanning five sub-domains. To our knowledge, this constitutes the first cross-domain co-occurrence analysis of XAI evaluation metrics in medical AI. We identified a dominant cluster of technically grounded metrics—Faithfulness and Sensitivity—that consistently co-occur. Perturbation-based metrics form a correlated secondary cluster ($\rho = 0.51$, $p < 0.001$), exhibiting partial methodological redundancy, while Sufficiency remains isolated (7.7%) despite direct relevance to clinical decision adequacy.

Mapping the corpus onto a three-layer taxonomy confirmed that evaluation is heavily concentrated in mathematical faithfulness (90.4%), with system robustness (34.6%) and human-centred trust (13.5%) substantially underrepresented—leaving the dimensions most consequential for clinical deployment systematically underserved.

These findings support two conclusions. First, a minimal evaluation framework—combining one faithfulness measure, one perturbation-based measure, one sensitivity metric, one sufficiency check, and at least one human-centred criterion—covers all three taxonomy layers without unreasonable methodological burden. Second, the field requires shared benchmarks and reporting standards, particularly for underserved sub-domains such as neuroimaging. Addressing these gaps is necessary for medical AI systems whose explanations are not only technically faithful but genuinely trustworthy in clinical practice.

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